

AADK 6 - Task 3 — Usability Review & Accessibility for Interactive Buttons

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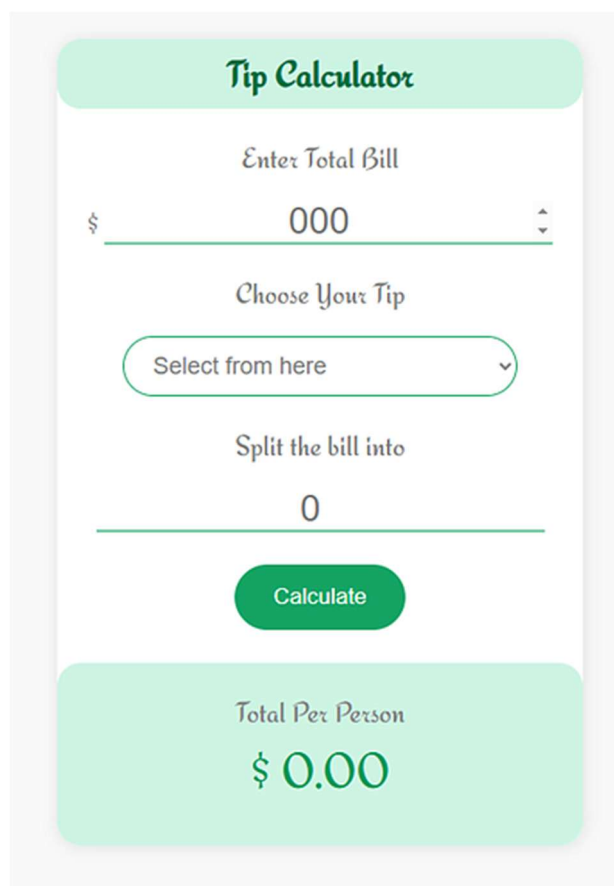
Course/Unit: Android Development with Kotlin – Session 6

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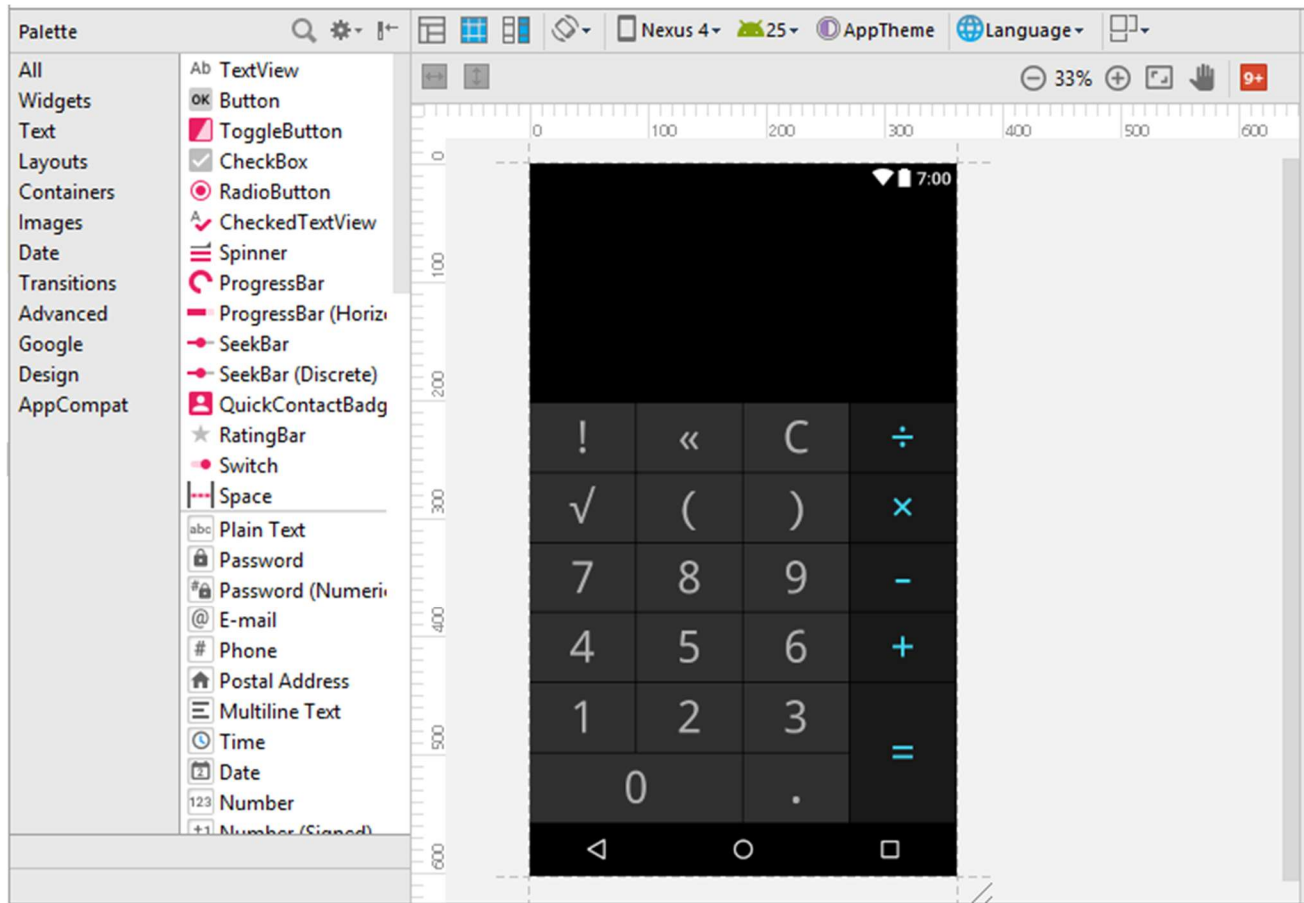
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Selected Screen

Example screen used for review: **Tip Calculator** screen with three buttons: **Calculate Tip**, **Split Bill**, and **Reset**.



The screenshot displays a mobile application titled "Tip Calculator". The interface includes a text input field for "Enter Total Bill" with a dollar sign prefix and a numeric value of "000". Below this is a "Choose Your Tip" section with a dropdown menu currently showing "Select from here". Further down is a "Split the bill into" section with a numeric input field showing "0". A green "Calculate" button is positioned below these inputs. At the bottom, a green box displays the result "Total Per Person" as "\$ 0.00".



1. Usability & Accessibility Checklist Review

Checklist Item	Observation
Button Size	Some buttons are smaller than recommended 48dp touch target .
Text Legibility	Text is readable but contrast between button background and text is slightly low.
Logical Grouping	Action buttons are placed together but spacing between them is inconsistent.
Feedback on Click	Buttons provide ripple feedback but no additional visual confirmation.
Accessibility Labels	Buttons do not have contentDescription for screen readers.
Adaptive Layout	On smaller screens buttons appear cramped and may overlap if more actions are added.

2. Review Table (Issues Identified)

Issue	Description	Proposed Fix	Priority
Small Touch Target	Buttons are smaller than 48dp , making them harder to tap.	Increase button height to 48dp+ and add padding.	High
Low Text Contrast	Text color and button background do not have strong contrast.	Use Material color contrast guidelines (e.g., white text on darker button).	High
Missing Accessibility Labels	Screen readers cannot clearly identify button actions.	Add <code>contentDescription</code> or semantic labels for each button.	High
Inconsistent Spacing	Buttons are too close to each other, increasing accidental taps.	Add 8–16dp spacing between buttons.	Medium
Limited Layout Responsiveness	Layout does not wrap properly on small devices.	Use Column/Row with wrap or responsive layout modifiers .	Medium

3. Example Accessibility Improvement (Jetpack Compose)

```
Button(  
    onClick = { calculateTip() },  
    modifier = Modifier  
        .fillMaxWidth()  
        .height(48.dp)  
        .padding(vertical = 8.dp)  
        .semantics { contentDescription = "Calculate tip button" }  
) {  
    Text(  
        text = "Calculate Tip",  
        fontSize = 16.sp  
    )  
}
```

}

Improvements:

- Ensures **48dp touch target**
 - Adds **screen reader accessibility**
 - Uses **adequate spacing**
-

4. Mini Report: Why usability and accessibility matter for interactive UI, especially buttons?

Usability and accessibility are essential for creating effective interactive UI elements. Buttons must be large enough to tap easily, clearly labelled, and provide visual feedback to guide users during interaction. Accessibility features like screen reader labels and proper colour contrast ensure that people with visual or motor impairments can use the app. Designing accessible buttons improves overall user experience and makes applications inclusive for all users.